

Contrasting use of hyporheic habitat by benthic invertebrates during spate and low flows

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Abstract

Spates are common disturbance events in lotic ecosystems, with high flows mobilising bed sediments and displacing biota. During spates, invertebrates inhabiting benthic sediments are typically exposed to higher flow velocities than those in the hyporheic zone, which may therefore act as a refugium. Between spates, declining discharge typically reduces the area of submerged sediments, concentrating populations of mobile taxa into a smaller inhabitable space. In such cases, the hyporheic zone has the potential to provide refuge from increasing biotic pressures in the surface stream. We compared the effects of a spate and flow recession on the abundance of benthic invertebrates in the hyporheic zone of a groundwater-dominated river. During flow recession, a reduction in the area of submerged sediments was associated with an increase in the population densities of the dominant amphipod *Gammarus pulex*. Subsequent increases in the hyporheic abundance of *G. pulex* and in the proportion of the total (benthic + hyporheic) population inhabiting the hyporheic zone indicated that these sediments acted as a refugium from biotic pressures in the benthic zone during low flows. During the spate, a decline in invertebrate abundance was most pronounced in the surface sediments, but significant reductions were also recorded in the hyporheic zone, indicating that the refugial capacity of these sediments was limited.

Introduction

Hydrological variability is central in structuring lotic ecosystems, and therefore directly influences the composition of instream invertebrate communities (Monk *et*

al., 2008). In particular, spates can restructure the streambed environment through sediment mobilisation (Olsen *et al.*, 2005) and high flows displace invertebrates, causing species richness and abundance to decline (Death, 2008). Such disturbance events can also maintain community diversity by reducing the dominance of competitive taxa (Death *et al.*, 1995).

During spates, increases in flow velocity and associated sediment mobilisation are typically less pronounced in the hyporheic zone than at the streambed surface, and these subsurface sediments therefore represent a potential refugium for benthic invertebrates (Williams *et al.*, 1974; Boulton *et al.*, 2010). However, evidence of this refugial role is ambiguous, with some studies demonstrating an increase in the hyporheic abundance of benthic invertebrates following a spate (Dole-Olivier *et al.*, 1997) and others noting no such increase (Boulton *et al.*, 2004). A low refugial capacity has been attributed to factors including the direction of hydrologic exchange (Dole-Olivier *et al.*, 1997) and disturbance magnitude being either too low (Boulton *et al.*, 2004) or too high (Hancock, 2006).

Between spates, flow recessions are accompanied by abiotic changes that can affect community composition (Dewson *et al.*, 2007). Hydrological changes typically include reduced flow velocity, depth and wetted width (McIntosh *et al.*, 2002), which may expose marginal habitats and mid-channel topographic high points (Dewson *et al.*, 2003). Contraction of the submerged area reduces habitat availability for benthic invertebrates, whilst the extent of the saturated hyporheic sediments remains constant until after surface water is lost (Griffith *et al.*, 1993). Hydrological changes during flow recession have secondary effects, for example dissolved oxygen may decline due to reduced mixing (Oliver *et al.*, 1999) and water chemistry may change in response to increased groundwater contributions (James *et al.*, 2008). However, low flows represent gradually changing and relatively moderate conditions compared with spate disturbances, and biotic forces (e.g. predation, competition) may therefore become more influential than physical forces in structuring communities (Franken *et al.*, 2006).

Whilst several studies have investigated the hyporheic zone refugium following streambed drying (e.g. Griffith *et al.*, 1993; Clinton *et al.*, 1996), few have considered use of these sediments by benthic invertebrates during low flows (James *et al.*, 2008; James *et al.*, 2009; Stubbington *et al.*, 2009). Of these studies, none has linked a decline in discharge to an increase in benthic invertebrate abundance in the hyporheic zone, despite predictions that abundances should rise if habitat contraction intensifies biotic pressures in the benthic sediments (James *et al.*, 2008).

Questions therefore remain regarding how benthic invertebrate use of hyporheic habitat changes in response to hydrological variability. In an effort to address this, this study recorded changes in benthic invertebrate abundance in the hyporheic zone of a groundwater-dominated river during a four-month flow recession then following a high-magnitude spate. This allowed comparison of hyporheic zone usage under contrasting hydrological conditions. It was predicted that: (i) benthic invertebrate use of the hyporheic zone would increase during the flow recession if benthic habitat availability declined; and (ii) that the hyporheic zone refugium would promote invertebrate survival during the spate.

Methods

Study area

The River Lathkill (Derbyshire, England; 53°11.2'N, 1°43.1'W) is a second-order stream flowing through a wooded valley. The region is temperate, with mean annual rainfall of 1,200 mm and a mean annual air temperature of 8.0 °C. The river is underlain by Carboniferous limestone and primarily discharges autogenic water (Banks *et al.*, 2009). Whilst the limestone aquifer provides significant baseflow, transmission losses occur through the streambed, primarily to underlying relic mining drainage levels. Consequently, partial or complete streambed drying occurs during the summer months. Four sites in the upper reaches were selected to represent the range of hydrological conditions experienced (Figure 1). The substrate at these sites consists of mixed alluvial deposits dominated by sand- to cobble-sized

clasts, and instream vegetation is primarily mosses and liverworts, with patches of reeds in marginal areas.

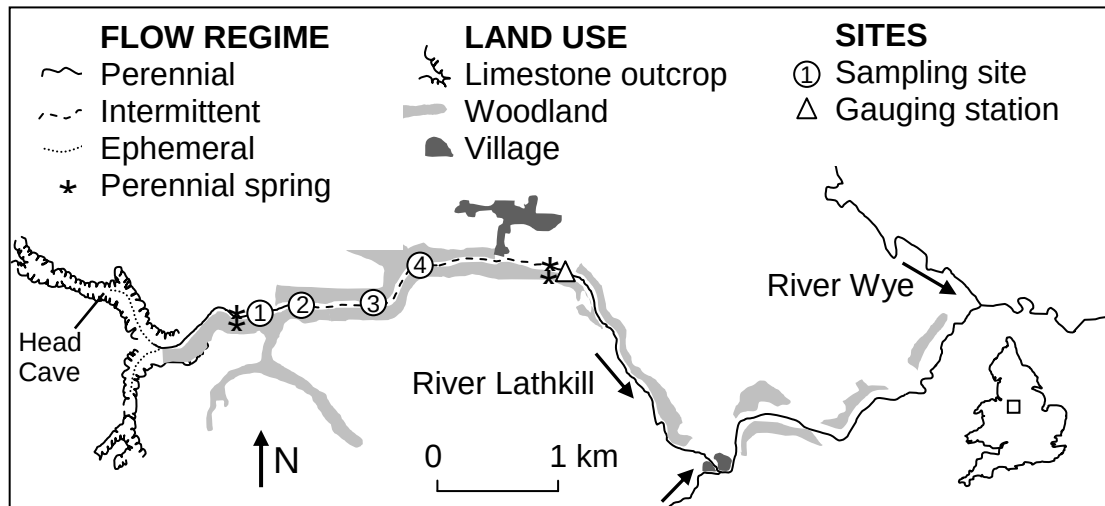


Figure 1. Location map of the River Lathkill.

Hydrological conditions

Discharge data was obtained from a gauging station approximately 1 km downstream of the study area, where discharge is augmented by perennial springs (Figure 1). Typically, flow recession begins in April, intermittent reaches dry by July, and discharge then increases in September. In 2008, discharge decreased sharply in early April then continued to decline gradually (Figure 2). When sampling commenced in mid-May, discharge was $0.43 \text{ m}^3 \text{ s}^{-1}$. Flow recession continued until mid-August, with the lowest mean daily discharges ($0.091\text{--}0.096 \text{ m}^3 \text{ s}^{-1}$) occurring between 5-16th August; sampling was conducted during this period, with partial streambed drying being apparent. A high-magnitude spate occurred in early September, with discharge increasing from $<0.23 \text{ m}^3 \text{ s}^{-1}$ to $>5.6 \text{ m}^3 \text{ s}^{-1}$ within 30

hours. September sampling was undertaken as flows receded, at a discharge of 0.7-1.1 m³ s⁻¹ (Figure 2).

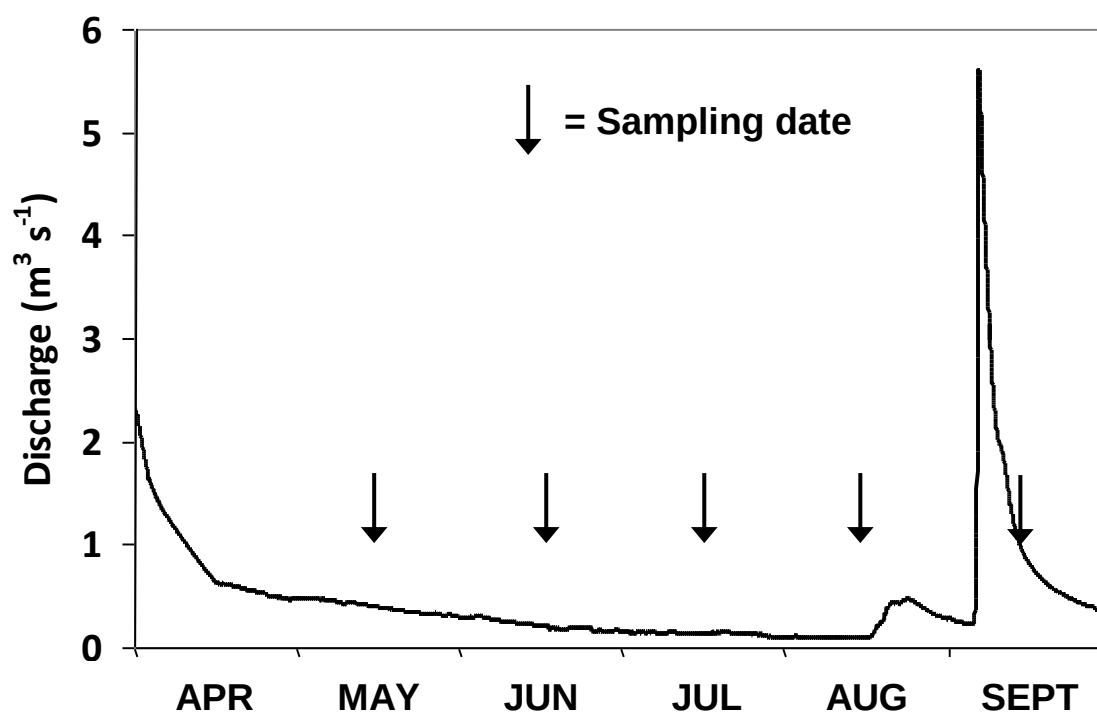


Figure 2. Hydrograph of mean hourly discharge (m³ s⁻¹) at a gauging station on the River Lathkill (see Figure 1 for location), April to September 2008.

Field Sampling

Sampling was undertaken at monthly intervals between May and September 2008. At each site, four sampling points were selected in riffle/run habitat. Prior to the first sampling date, three polyvinylchloride pipes (19 mm internal diameter) were installed at each sampling point to depths of 10, 20 and 30 cm respectively. These pipes functioned as permanent hyporheic invertebrate sampling wells and were sealed between sampling occasions to prevent colonisation by benthic invertebrates.

Each month, at each sampling point, benthic invertebrates were collected using a Surber sampler (0.1 m², 250 µm mesh net), by manually disturbing the substrate within the frame to a depth of 50 mm for 30 seconds. Large clasts were inspected and attached invertebrates included in the sample. Hyporheic invertebrates were

pumped from the pre-installed sampling wells using a hand-operated bilge pump (Boulton *et al.*, 1992). Each sample consisted of 6 litres of water, sediment and invertebrates, which was passed through a 125 µm sieve to retain invertebrates. All invertebrate samples were preserved using 4 % formalin.

Three hydrological variables were measured at each sampling point: water depth, flow velocity at 0.6 x depth and wetted width. Depth and velocity were measured using an ADS SENSAR-C2 flow meter (ADS Environmental Services, Huntsville, USA). Temperature, pH, conductivity and dissolved oxygen (DO; mg L⁻¹ and % saturation) were measured *in situ* in surface water and hyporheic water from each depth using standard instrumentation (Hanna Instruments, Leighton Buzzard, UK).

Laboratory analyses

Invertebrates were identified to the lowest taxonomic resolution possible, in many cases species level. However, *Baetis* (Ephemeroptera), some *Leuctra* and *Nemoura* (Plecoptera), *Oulimnius* and *Hydraena* (Coleoptera), Sphaeriidae (Bivalvia), larval Dytiscidae and Scirtidae (Coleoptera), some Diptera (Ceratopogonidae, Chironomidae, Empididae, Muscidae, Stratiomyidae, Simuliidae, Tipulidae), Cladocera, Cyclopoida, Harpacticoida, Nematoda, Oligochaeta and Hydracarina were left at the stated taxonomic levels.

Statistical analysis

Preliminary analysis using two-way repeated measures (RM) ANOVA with month as a within-subject variable and hyporheic depth (10, 20, 30 cm) as a between-subject factor indicated no significant differences between depths. Samples from all hyporheic depths were therefore combined prior to subsequent analysis of temporal change.

One-way RM ANOVA with month as a within-subject variable was used to assess temporal change in environmental parameters, with *post-hoc* paired *t*-tests with Bonferroni corrections used to determine where significant differences occurred.

For each month, total invertebrate abundance (TIA), number of taxa, Simpson's diversity and Berger-Parker dominance (a measure of the proportion of the community accounted for by the most common taxon) were calculated to characterise the benthic and hyporheic communities. Monthly abundance of dominant taxa (calculated separately for benthic and hyporheic samples and defined as taxa accounting for >1 % of all individuals) was also determined. To assess the strength of interactions between benthic and hyporheic invertebrate communities, the proportion of the total (benthic + hyporheic) community found within the hyporheic zone was calculated for TIA and selected benthic taxa. Using proportional data allows comparison of populations sampled using different techniques.

Indices, abundances, and hyporheic proportions were used as dependent variables in subsequent analysis of temporal change. Prior to analysis, abundance data were square-root transformed and proportional data were arcsine square-root transformed to reduce skewness and moderate the influence of dominant taxa (Underwood, 1997). One-way RM ANOVA was then used to consider patterns of temporal change in biotic variables, as outlined for environmental variables.

Results

Environmental conditions

Both surface water depth and wetted width declined between May and June, increased slightly in July, declined to the lowest recorded levels in August, then peaked in September after the spate ($p \leq 0.002$; Table 1). Flow velocities declined gradually between May and July, increased slightly in August then rose sharply in September ($p < 0.001$); measured post-flood peak flow velocities reached 1.16 m s^{-1} .

Exposure of some hyporheic sampling wells in September indicated that sediment scour of ≤ 20 cm had occurred during the spate.

Temporal change was significant for DO (mg L^{-1}), conductivity, pH and temperature, although conductivity, pH and temperature remained moderate throughout the study (Table 1). DO concentrations were significantly higher in surface water compared with hyporheic water ($p < 0.001$) and declined between June and August in both environments ($p \leq 0.035$; insufficient May and September data available for inclusion in analysis; Table 1).

Table 1: Temporal change in hydrological and water chemistry variables on the River Lathkill, May to September 2008.

Values are presented as the mean $\pm$ 1 SE of all samples. Each month, $n = 16$ for surface water and $n = 48$ for hyporheic water (combined data from depths of 10, 20 and 30 cm), except for dissolved oxygen data collected in May, where $n = 5$ for surface water and $n = 15$ for hyporheic water. Temporal change analysed using one-way RM ANOVA; * indicates $p < 0.01$.

		<i>Surface/ Hyporheic</i>	<i>May</i>	<i>June</i>	<i>July</i>	<i>Aug</i>	<i>Sept</i>	<i>Temporal change</i>
Hydrological variables	Water depth (cm)	surface	16.6 $\pm$ 1.59	8.4 $\pm$ 1.24	10.1 $\pm$ 1.35	5.9 $\pm$ 1.09	30.3 $\pm$ 2.5	*
	Velocity (m s ⁻¹)	surface	0.29 $\pm$ 0.04	0.21 $\pm$ 0.04	0.12 $\pm$ 0.02	0.16 $\pm$ 0.04	0.56 $\pm$ 0.06	*
	Wetted width (m)	surface	8.1 $\pm$ 1.3	6.2 $\pm$ 1.0	6.6 $\pm$ 1.0	5.9 $\pm$ 1.0	9.1 $\pm$ 1.1	*
Water chemistry	Dissolved oxygen (mg L ⁻¹)	surface	12.5 $\pm$ 0.35	9.7 $\pm$ 0.26	10.3 $\pm$ 0.83	8.4 $\pm$ 0.46	not available	*
		hyporheic	7.8 $\pm$ 0.4	6.7 $\pm$ 0.2	6.8 $\pm$ 0.4	5.2 $\pm$ 0.3	not available	*
	Dissolved oxygen (% sat.)	surface	100 $\pm$ 0	94.3 $\pm$ 1.3	92.2 $\pm$ 2.2	89.1 $\pm$ 2.7	not available	-
		hyporheic	85.5 $\pm$ 3.0	74.0 $\pm$ 1.7	75.8 $\pm$ 2.9	69.4 $\pm$ 2.3	not available	-
	Temperature (°C)	surface	11.1 $\pm$ 0.4	10.0 $\pm$ 0.2	10.8 $\pm$ 0.3	10.3 $\pm$ 0.3	9.4 $\pm$ 0.1	*
		hyporheic	11.0 $\pm$ 0.2	9.9 $\pm$ 0.1	11.4 $\pm$ 0.1	10.8 $\pm$ 0.2	9.5 $\pm$ 0.1	*
	pH	surface	8.1 $\pm$ 0.07	8.2 $\pm$ 0.04	8.4 $\pm$ 0.10	7.6 $\pm$ 0.06	7.8 $\pm$ 0.03	*
		hyporheic	8.0 $\pm$ 0.03	8.1 $\pm$ 0.05	8.3 $\pm$ 0.06	7.1 $\pm$ 0.03	7.8 $\pm$ 0.03	*
	Conductivity (μS cm ⁻¹)	surface	579 $\pm$ 1.9	599 $\pm$ 2.7	617 $\pm$ 5.1	619 $\pm$ 5.9	597 $\pm$ 2.5	*
		hyporheic	590 $\pm$ 1.0	611 $\pm$ 1.5	624 $\pm$ 2.1	629 $\pm$ 3.3	607 $\pm$ 0.6	*

Invertebrate communities

In total, 31,415 individuals from 63 taxa were recorded from 82 Surber samples. *Gammarus pulex* (Crustacea: Amphipoda) dominated the benthic community, accounting for 38.9 % of total invertebrate abundance (TIA). The Chironomidae (Diptera) were also abundant, comprising 18.7 % TIA. Ten other taxa made up 1-10 % TIA: the flatworm *Polycelis felina*, the Oligochaeta, two mayflies (*Serratella ignita*, *Baetis* spp.), a stonefly (*Leuctra* spp.), two caddisflies (*Agapetus fuscipes*, *Drusus annulatus*), two riffle beetle larvae (*Elmis aenea*, *Riolus subviolaceus*) and the family Psychodidae (Diptera).

A total of 7562 invertebrates from 52 taxa were recorded in 240 hyporheic samples from depths of 10-30 cm. The most common hyporheic taxa were the Chironomidae and the Ostracoda (Crustacea), comprising 25.9 % and 14.7 % TIA respectively. The predominantly benthic taxa *Polycelis felina* and *G. pulex* were common in the hyporheic zone, accounting for 13.8 % and 13.4 % TIA respectively. Another six taxa comprised 1-10 % of the hyporheic fauna: *Leuctra* spp., *Baetis* spp., *Nemoura* spp., Cyclopoida, Oligochaeta and Nematoda.

Temporal change in benthic community composition

TIA increased moderately between May and June, remained stable between June and August, then declined sharply in September after the spate ($p = 0.004$; Figure 3a). In contrast, *G. pulex* abundance increased threefold between May (78 individuals (ind.) 0.1 m^{-2}) and August (257 ind. 0.1 m^{-2} ; $p = 0.011$) with peak abundances of 648 ind. 0.1 m^{-2} occurring in August; abundance then declined by 79.8 % in September ($p < 0.001$; Figure 3b). Other significant temporal changes in the abundance of dominant taxa were restricted to the Insecta; such changes may reflect seasonal emergence of adults and are not considered further. Variation in *G. pulex* abundance was accompanied by changes in the proportion of TIA accounted for by this species, which rose from 0.27 in May to 0.48 in August then returned to 0.27 in September (Figure 3c); this was reflected by significant changes in Berger-

Parker dominance, which increased steadily between May and August then declined in September ($p < 0.001$; Figure 3d). The opposite pattern was observed for Simpson's diversity, which decreased between May and August before recovering in September ($p < 0.001$; Figure 3e). Taxon richness was highest in May and June, declined in July and fell again in September ($p = 0.003$; Figure 3f).

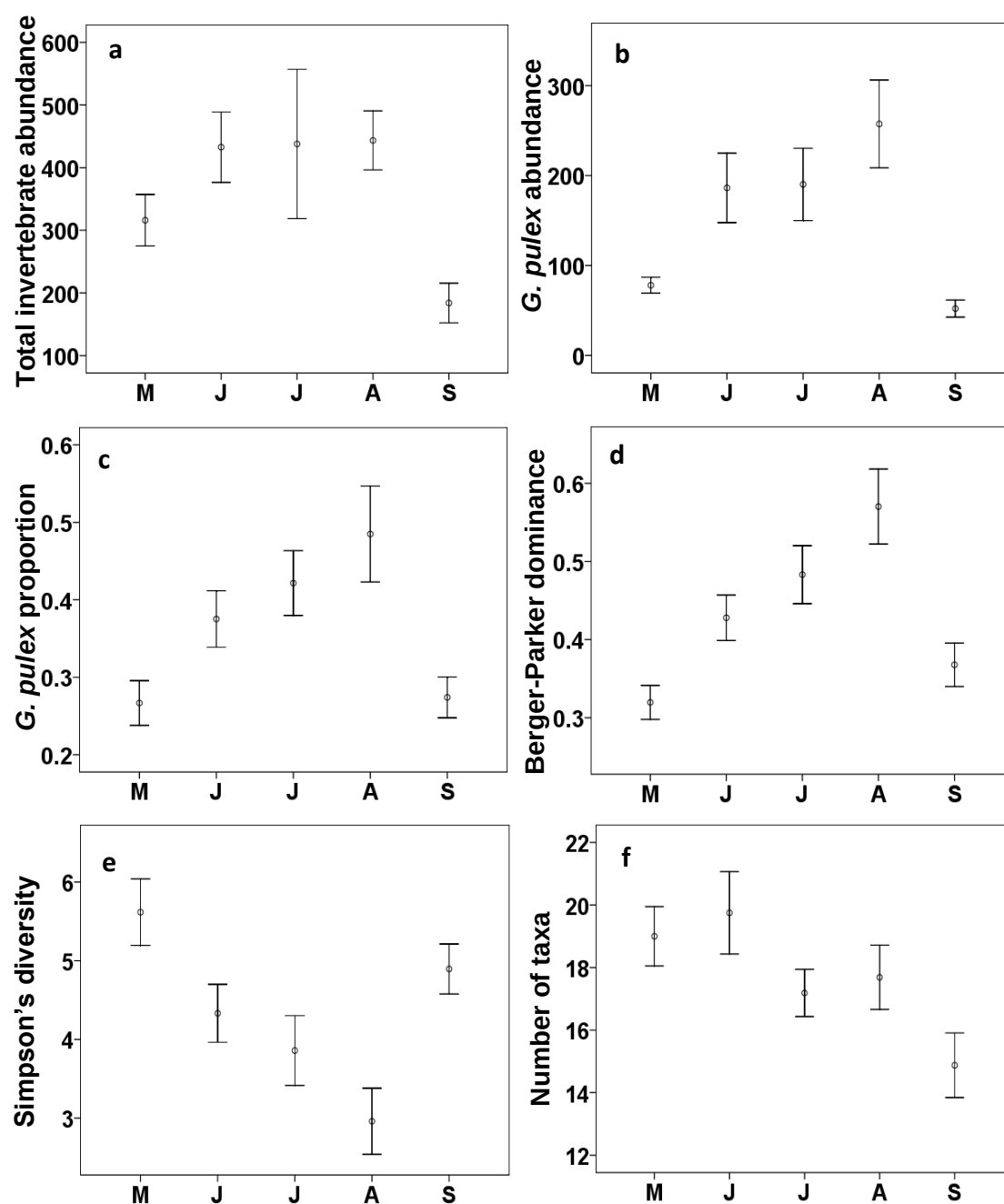


Figure 3. Mean ± 1 SE benthic community composition, May to September 2008: a) total invertebrate abundance (TIA; individuals 0.1 m^{-2}); b) abundance of *Gammarus pulex* (individuals 0.1 m^{-2}); c) *G. pulex* abundance as a proportion of TIA; d) Berger-Parker dominance index; e) Simpson's diversity index; f) number of taxa (0.1 m^{-2}).

Occurrence of benthic invertebrates in the hyporheic zone

Two primarily benthic taxa, *G. pulex* and *P. felina*, were common in the hyporheic community; consideration of temporal change in hyporheic zone use by benthic taxa is therefore restricted to these species. *G. pulex* abundance was similar and low in May and June, increased threefold between June and August then returned to May/June levels in September ($p < 0.001$; Figure 4a). *P. felina* abundance was also low in May and June, increased sharply between June and August then decreased moderately in September ($p = 0.002$; Figure 4b). These changes in abundance did not significantly alter the proportion of hyporheic TIA accounted for by either individual taxon ($p \geq 0.126$). However, when *G. pulex* and *P. felina* abundances were combined into a single metric describing the benthic component of the hyporheic community, this proportion was higher in August than other months ($p = 0.002$; Figure 5).

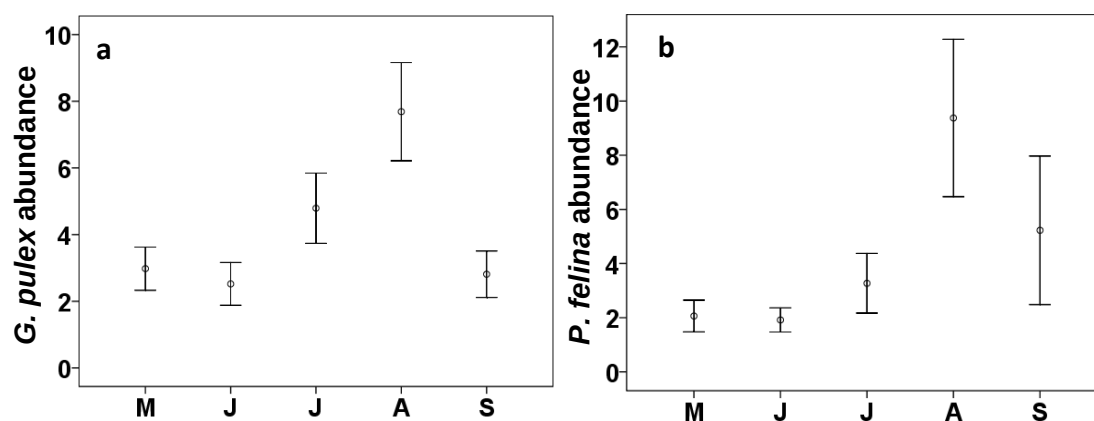


Figure 4: Mean $\pm$ 1 SE abundance (individuals 6 L⁻¹) of benthic invertebrates in the hyporheic zone: a) *Gammarus pulex*; b) *Polycelis felina*.

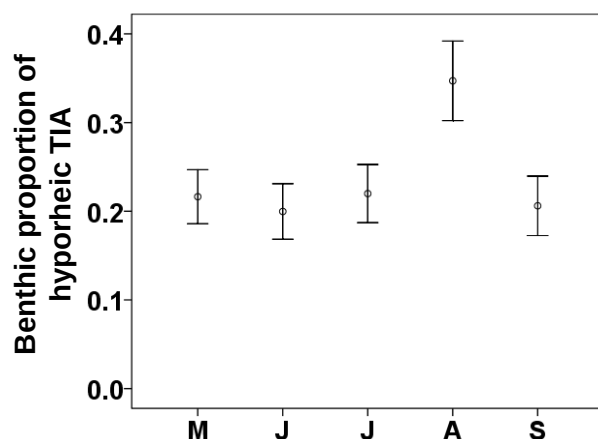


Figure 5: Mean $\pm$ 1 SE abundance of benthic taxa (*Gammarus pulex* and *Polycelis felina*) as a proportion of all invertebrates in the hyporheic zone (hyporheic TIA).

The hyporheic proportion of the total invertebrate community

The hyporheic proportion of the total (benthic + hyporheic) invertebrate community was lowest in May and June, intermediate in July and August and peaked in September ($p = 0.006$; Figure 6a). The proportion of the *G. pulex* community inhabiting the hyporheic zone declined slightly between May and June then increased threefold to an August peak and remained high in September; these changes were not significant ($p = 0.132$; Figure 6b). The hyporheic proportion of *P. felina* varied moderately between 0.32 in June/July and 0.58 in August ($p = 0.211$).

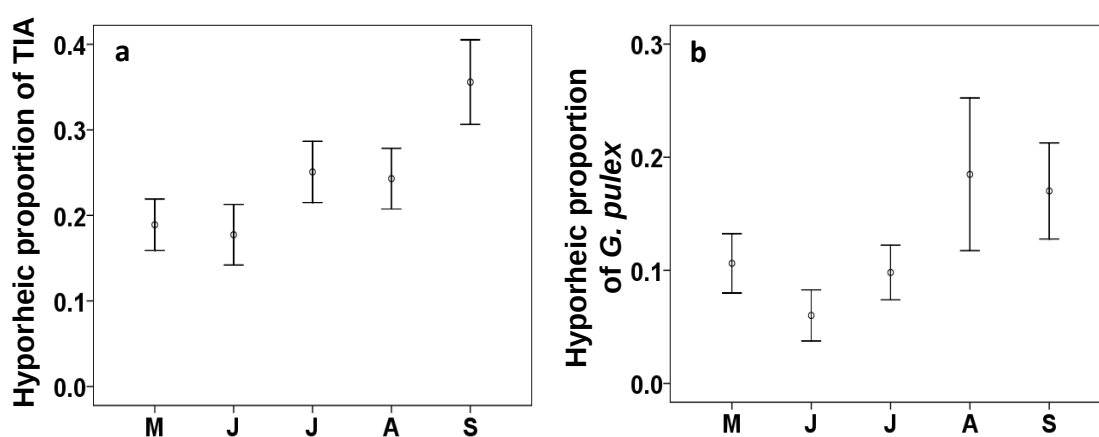


Figure 6: Mean $\pm$ 1 SE proportion of the total (benthic + hyporheic) invertebrate community inhabiting the hyporheic zone: a) total invertebrate abundance (TIA); b) *Gammarus pulex* abundance.

Discussion

Environmental conditions

The flow recession represented a prolonged period of moderate instream conditions; nonetheless, low flows can have detrimental effects on invertebrate communities by reducing habitat availability and water quality (Dewson *et al.*, 2007). In the current study, most water chemistry parameters experienced only minor changes which were unlikely to have had biotic effects. DO concentrations declined between June and August, but mean % saturation remained >90 % in surface water, which is 'excellent' in terms of ecological health (Gordon *et al.*, 2004). It is therefore probable that the primary controls on the benthic community were hydrological: declining depth, width and velocity, and therefore reduced habitat availability and heterogeneity during the flow recession, and high flow velocities capable of mobilising sediment and causing catastrophic invertebrate drift during the spate (Death, 2008).

The principal difference between surface and hyporheic water chemistry was lower DO availability in the latter, with the lowest values coinciding with lowest flows. However, mean concentrations remained >5 mg L⁻¹, which both *G. pulex* and *P. felina* can tolerate (Russier-Delolme, 1974; Grant *et al.*, 1982); the hydrological habitat provided by the hyporheic zone therefore appears to have remained adequate for invertebrate fauna throughout the study.

Invertebrate community change in the surface sediment

Whilst benthic TIA remained stable during flow recession, *Gammarus pulex* abundance trebled, causing community diversity to decline. Increased *G. pulex* population densities could either reflect a numerical increase in abundance or the concentration of individuals into a contracting habitat. Changes in *G. pulex* abundance closely reflected habitat availability, suggesting the latter scenario. In either case, *G. pulex* is a mobile species with multiple modes of feeding including

shredding of organic matter, predation and cannibalism (Kelly *et al.*, 2002) and rising population densities are likely to have increased biotic stresses in the benthic sediments.

Between August and September, benthic TIA decreased by 59.0 %, whilst *G. pulex* abundance experienced particularly pronounced declines of 79.8 %. These reductions undoubtedly reflect catastrophic drift during the spate, and were important in reducing *G. pulex* dominance and restoring community diversity (cf. Death *et al.*, 1995).

Benthic invertebrate use of the hyporheic zone

The increase in the hyporheic abundance of both *G. pulex* and *P. felina* did not commence until July, a month after benthic population densities began rising. This suggested that migration of benthic invertebrates into the hyporheic zone occurred in response to the postulated increase in biotic pressures in the surface sediments. This assertion is supported by experimental studies reporting increased use of interstitial habitat by *G. pulex* in response to cannibalistic and predation pressures (Franken *et al.*, 2006; McGrath *et al.*, 2007). A refugium effect is also suggested by moderate increases in the hyporheic proportions of the total (benthic + hyporheic) *G. pulex* and *P. felina* populations; that these increases were not significant may reflect numerous seasonal changes in the abundance of insect taxa (data not shown).

In contrast, the abundance of both *G. pulex* and *P. felina* declined in the hyporheic zone between August and September, probably due to sediment scour during the spate, although % reductions were lower than in the benthic sediments.

Some previous studies have recorded actual increases in benthic invertebrate abundance in the hyporheic zone after a disturbance, providing evidence of *active* refugium use, i.e. increased survival resulting from active vertical migrations (Williams *et al.*, 1974; Clinton *et al.*, 1996). In the current study, changes in the

hyporheic abundance of both *G. pulex* and *P. felina* provide evidence of active refugium use during flow recession, but not the spate. In contrast, other previous research has cited increases in *relative* abundance of benthic invertebrates or simply the presence of benthic taxa in the hyporheic zone as evidence of *passive* refugium use, i.e. invertebrates already inhabiting hyporheic sediments experience greater survival than those in the surface sediments (Griffith *et al.*, 1993; Fenoglio *et al.* 2006). In the current study, the hyporheic zone acted as a passive refugium during both the flow recession and the spate. This study therefore provides evidence that both active and passive refugium use promote benthic invertebrate survival and that the importance of these modes of refugium use depends on the nature of the disturbing forces.

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